Chapter 5, Section 3

James Lee

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1. Let X be a nonsingular projective variety of any dimension, let Y be a nonsingular subvariety, and let $\pi : \tilde{X} \to X$ be obtained by blowing up Y. Show that $p_a(\tilde{X}) = p_a(X)$.

2. Let C and D be curves on a surface X, meeting at a point P. Let $\pi : \tilde{X} \to X$ be the monoidal transformation with center P. Show that $\tilde{C}.\tilde{D} = C.D - \mu_P(C) \cdot \mu_P(D)$. Conclude that $C.D. = \sum \mu_P(C) \cdot \mu_P(D)$, where the sum is taken over all intersection points of C and D, including infinitely near intersection points.

3. Let $\pi : \tilde{X} \to X$ be a monoidal transformation, and let D be a very ample divisor on X. Show that $2\pi^* D - E$ is ample on $\tilde{X}$.

4. *Multiplicity of a Local Ring.* Let A be a Noetherian local ring with maximal ideal $\mathfrak{m}$. For any $l > 0$, let $\psi(l) = \text{length}(A/\mathfrak{m}^l)$. We call ψ the *Hilbert-Samuel function* of A.

(a) Show that there is a polynomial $P_A(z) \in \mathbf{Q}[z]$ such that $P_A(l) = \psi(l)$ for all $l \gg 0$. This is the *Hilbert-Samuel polynomial* of A.

(b) Show that $\deg P_A = \dim A$.

(c) Let $n = \dim A$. Then we define the *multiplicity* of A, denoted $\mu(A)$, to be $(n!) \cdot (\text{leading coefficient of } P_A)$. If P is a point on a Noetherian scheme X, we define the *multiplicity* of P on X, $\mu_P(X)$, to be $\mu(\mathcal{O}_{P,X})$.

(d) Show that for a point P on a curve C on a surface X, this definition of $\mu_P(C)$ coincides with the one in the text just before (3.5.2).

(e) If Y is a variety of degree d in $\mathbf{P}^n$, show that the vertex of the cone over Y is a point of multiplicity d.

5. Let $a_1, \dots, a_r$, $r \geq 5$, be distinct elements of k, and let C be the curve in $\mathbf{P}^2$ given by the (affine) equation $y^2 = \prod_{i=1}^r (x - a_i)$. Show that the point P at infinity on the y-axis is a singular point. Compute δ_P and $g(\tilde{Y})$, where $\tilde{Y}$ is the normalization of Y. Show in this way that one obtains hyperelliptic curves of ever genus $g \geq 2$.

6. Show that analytically isomorphic curve singularities (I, 5.6.1) are equivalent in the sense of (3.9.4), but not conversely.

7. For each of the following singularities at $(0, 0)$ in the plane, give an embedded resolution, compute δ_P, and decide which ones are equivalent.

(a) $x^3 + y^5 = 0$

(b) $x^3 + x^4 + y^5 = 0$

(c) $x^3 + y^4 + y^5 = 0$

(d) $x^3 + y^5 + y^6 = 0$

(e) $x^3 + xy^3 + y^5 = 0$

8. Show that the following two singularities have the same multiplicity, and the same configuration of infinitely near singular points with the same multiplicities, hence the same δ_P, but are not equivalent.

(a) $x^4 - xy^4 = 0$

(b) $x^4 - x^2y^3 - x^2y^5 + y^8 = 0$